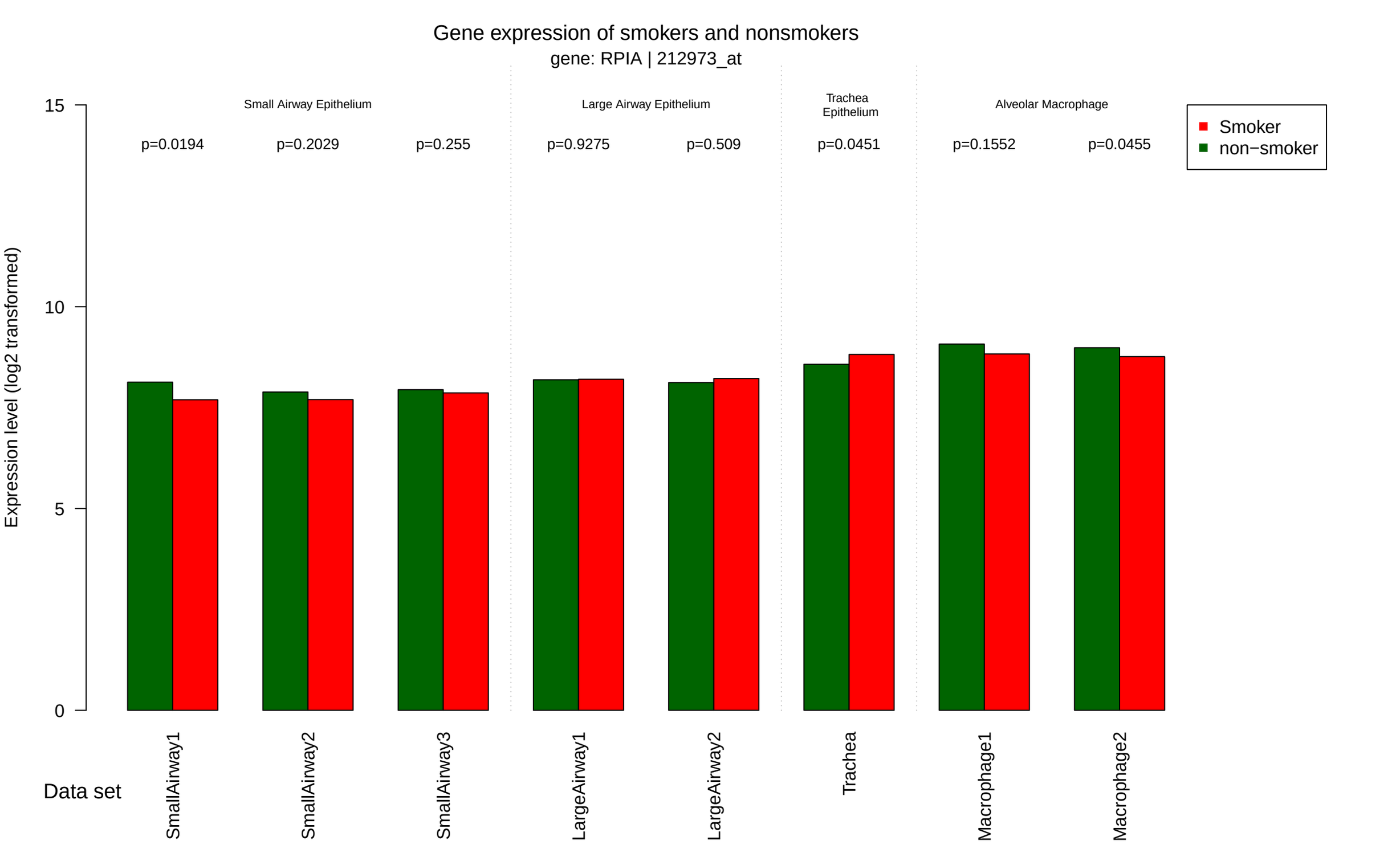
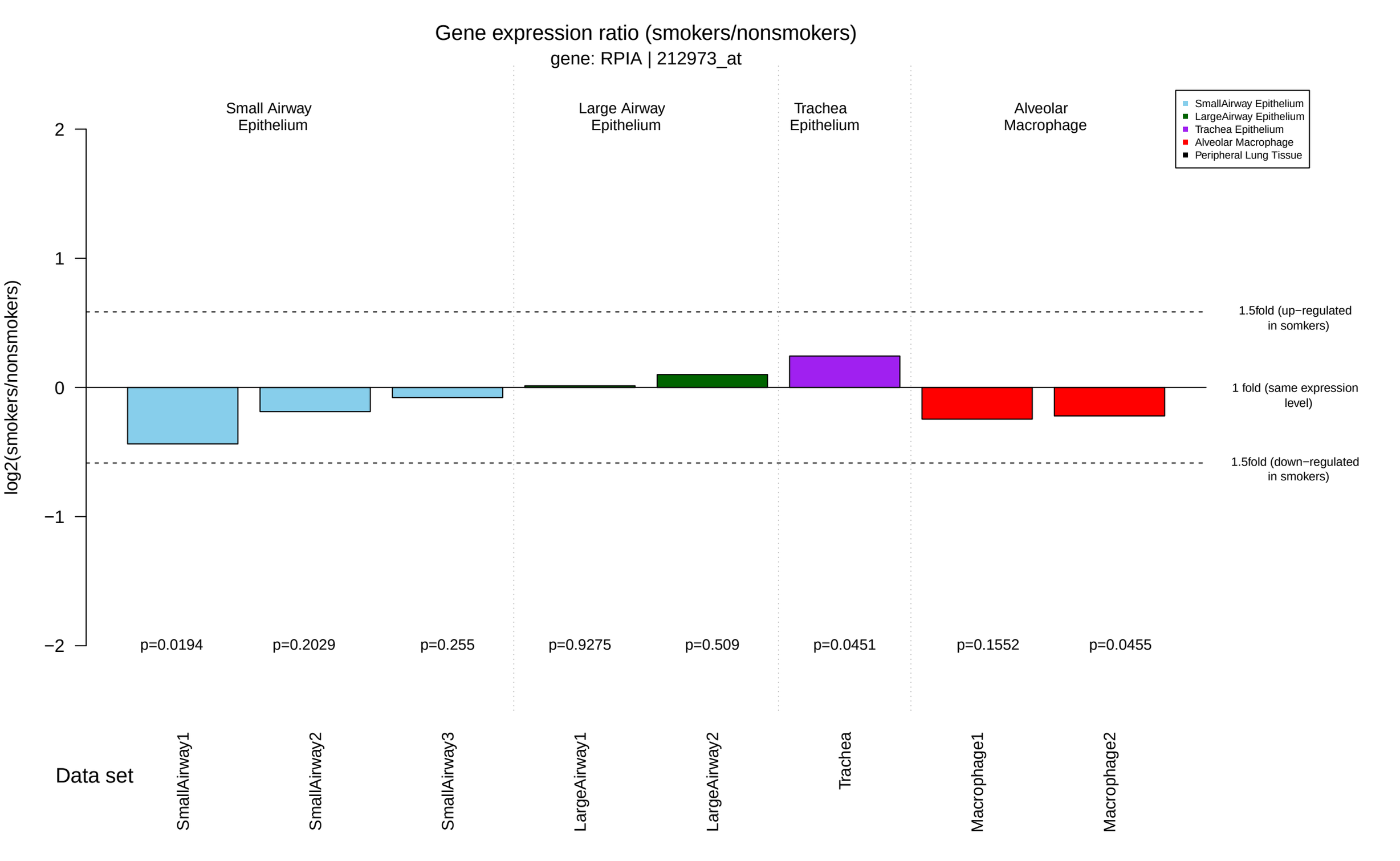
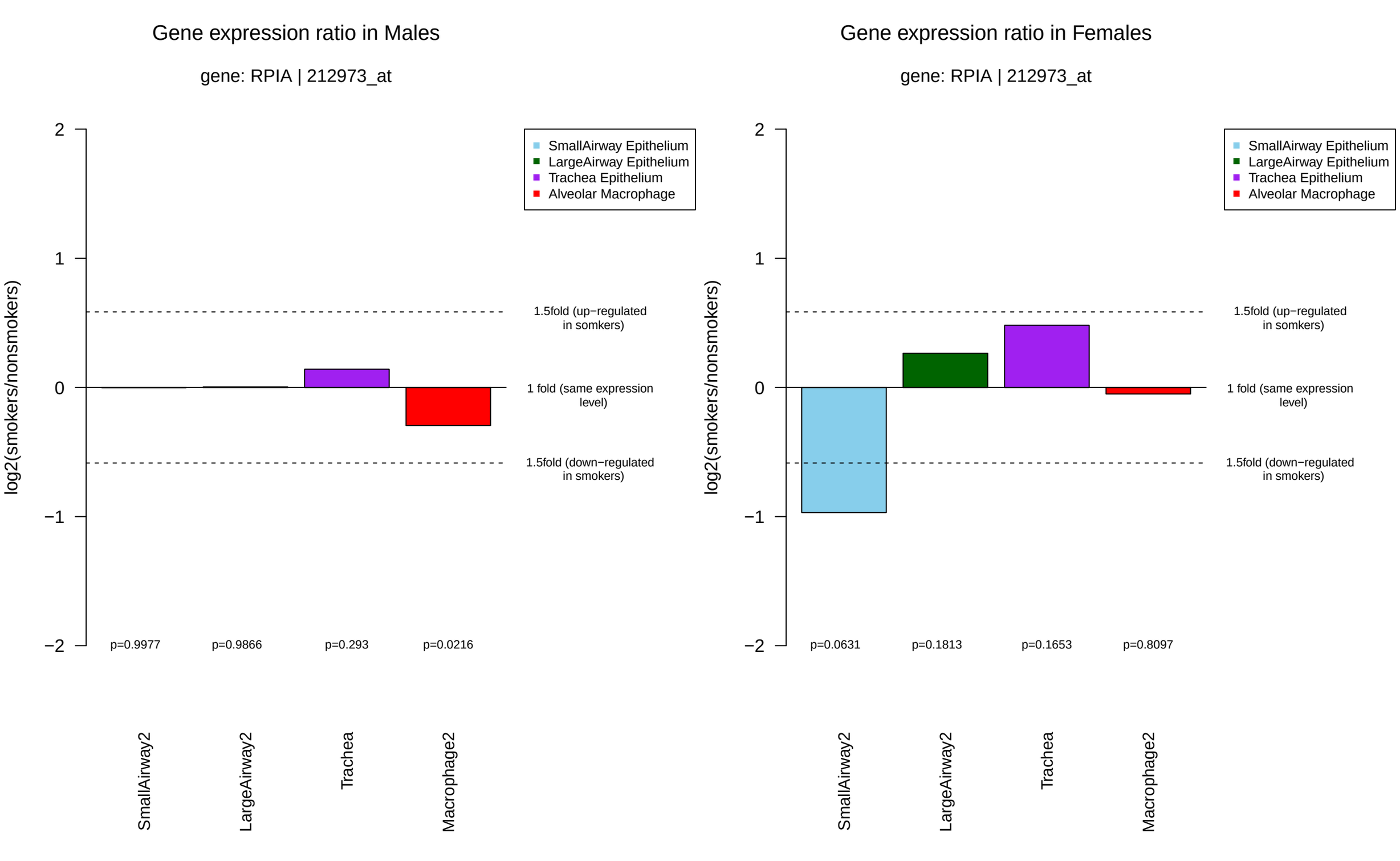
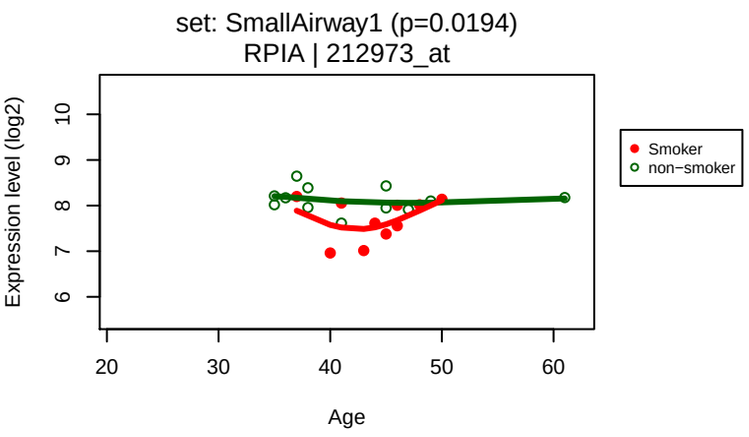
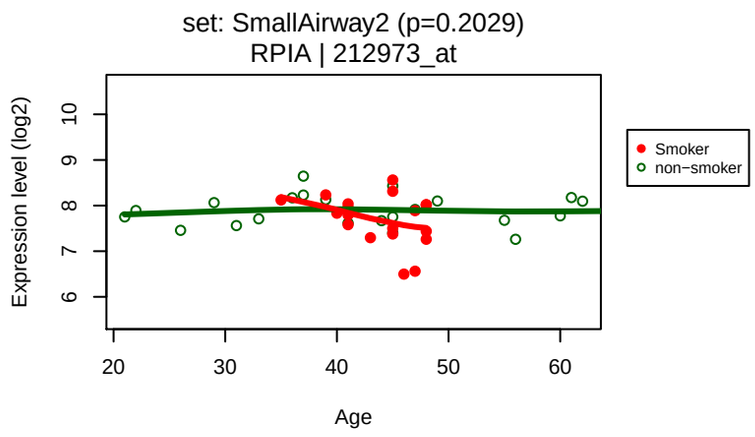
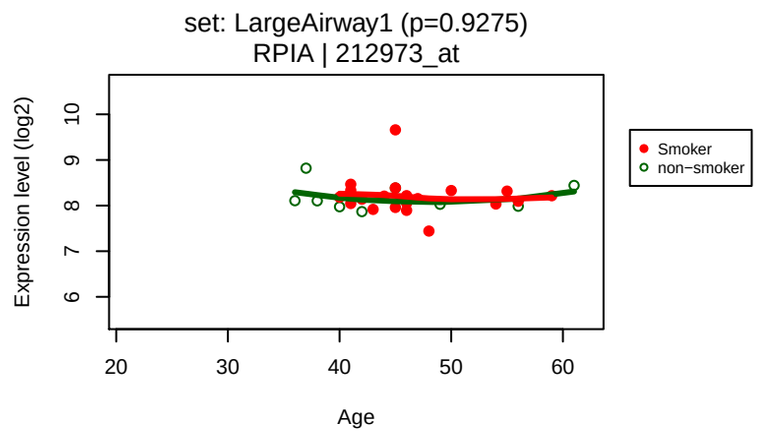

Gene expression of smokers and nonsmokers
gene: RPIA | 212973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0194
p=0.2029
p=0.255
p=0.9275
p=0.509
p=0.0451
p=0.1552
p=0.0455
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

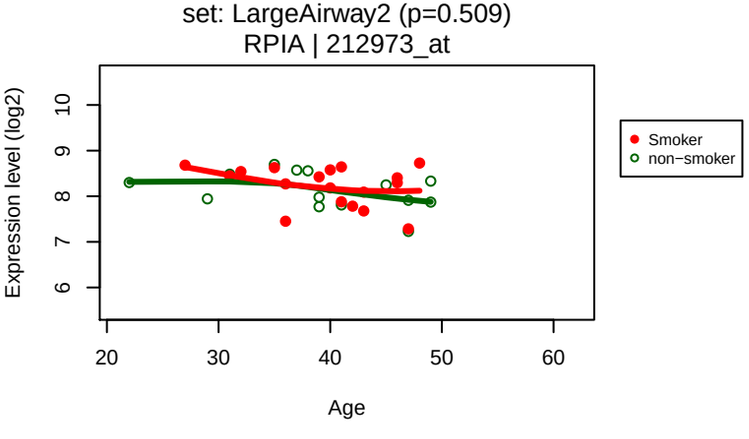

Gene expression ratio (smokers/nonsmokers)
gene: RPIA | 212973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0194
p=0.2029
p=0.255
p=0.9275
p=0.509
p=0.0451
p=0.1552
p=0.0455
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

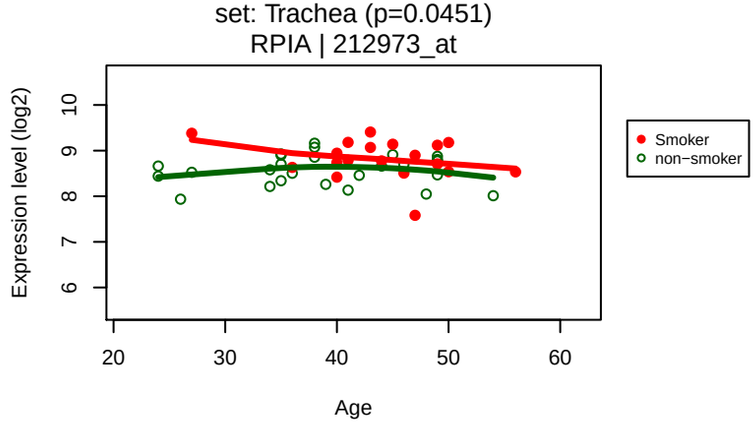

Gene expression ratio in Males
gene: RPIA | 212973_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9977
p=0.9866
p=0.293
p=0.0216
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPIA | 212973_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0631
p=0.1813
p=0.1653
p=0.8097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

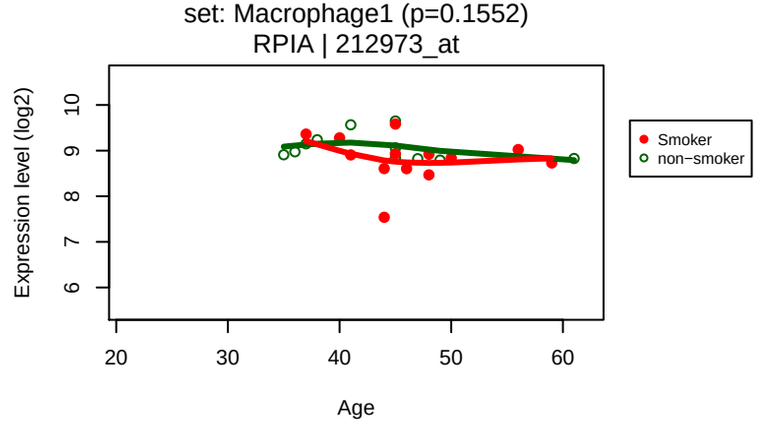

set: SmallAirway1 (p=0.0194)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker

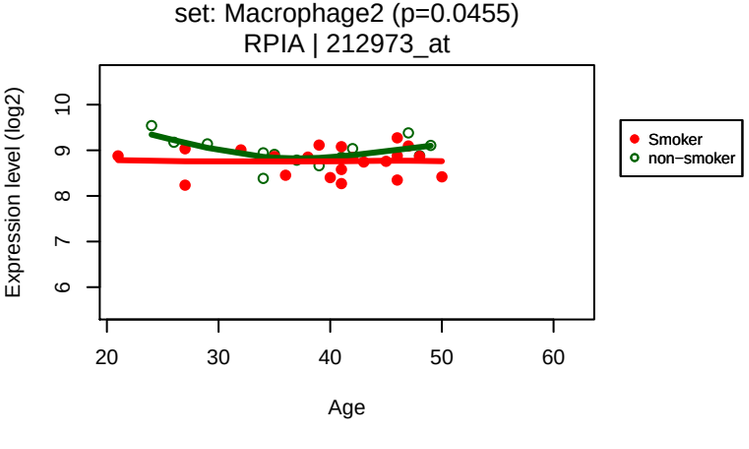

set: SmallAirway2 (p=0.2029)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker

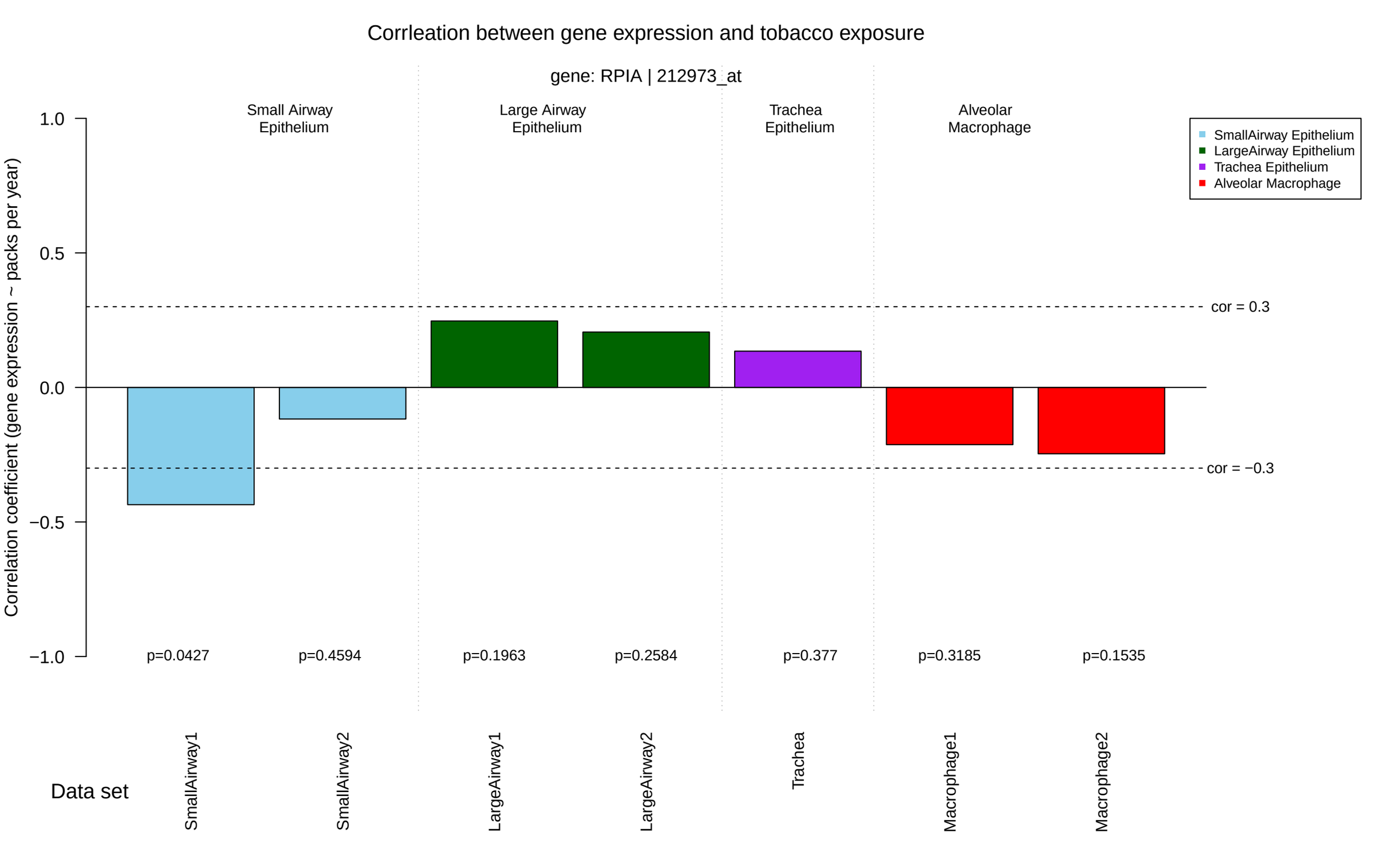

set: LargeAirway1 (p=0.9275)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.509)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0451)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1552)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0455)
RPIA | 212973_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPIA | 212973_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0427
p=0.4594
p=0.1963
p=0.2584
p=0.377
p=0.3185
p=0.1535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2